Design and Analysis of Algorithms

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Contact Details

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  - Office Hours : Will be Displayed outside my office.
- **Schedule**
  - 2 Classes of 1.5 hours

Design and Analysis of Algorithms

- Online course content & coordination
  - Google Classroom

Text book and reference material

- Introduction to Algorithms (Text Book)

  Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein

  Third Edition, MIT Press
- Any web material (consult authentic material e.g. on some university’s website)

Grading Criteria (Marks Distribution)

- Tentative grading criteria is as follows:
  - Quizzes (10%)
  - Assignments (10%)
  - Project (10%)
  - Mid Term Exam (30%)
  - Final Exam (40%)

Grading Policy

- There is simply no chance of extension in any of the deadline, what so ever
- You can request for re-checking of any of your evaluation as per following rules;

  Exams: Same day

  Assignments: 2 days after handing-over

  Quizzes: 2 days after handing over

  Warning! After due time, request will not be considered even if it's genuine.

  Warning! Keep checking your FLEX regularly and don't come in the end with bulk of queries in hand which you never presented earlier and now you are on the Edge.

General Guidelines

- Cheating cases are intolerable. You will be given negative marks for cheated stuff irrespective of the fact that, you were provider or the other one. Your cheating in exam will make it easy for you to step down from the Course with an **‘F’ grade**. . . ☹
- Plagiarism in one item of the assessment instrument will result in cancellation of all items of corresponding instrument.
- There will be **NO RE-TAKE** of any Quiz, Assignment and Project.
- Quiz is inevitable so always, expect a One ☺ [at least, one in every week]

General Guidelines

- You have to depend on yourself to have a good grade in the course
  - *Written Assignment assessment is based on the quizzes (60/40 ratio). In case of missed quiz for the assignment, 40% marks will be deducted from the assignment (if it is not*
  - *plagiarized)*
  - *Grading will be individual even for group tasks. So better to shine with your own work in hand otherwise don't complaint*

Algorithm

- An algorithm is a well-defined and effective sequence of computation steps that takes some value, or set of values, as input and produces some value, or set of values, as output.

Questions?

- What are algorithms?
- Why is the study of algorithms worthwhile?
- What is the role of algorithms relative to other technologies used in computers?

Example: sorting

- Input: A sequence of n numbers <a1,a2,a3…an>
- Output: A permutation (re-ordering) <b1,b2,b3…bn> of the input sequence such that b1<b2<b3…<bn

Correctness of an algorithm

- An algorithm is said to be correct if, for every **input instance**, it halts with the correct output.
- An incorrect algorithm
  - might not halt at all on some input instances, or
  - It might halt with an answer other than desired one.

Problems solved by algorithms

- Sorting/searching are by no mean the only computational problem for which algorithms have been developed.
- Otherwise, we wouldn't have the whole course on this topic
- Practical application of algorithms are <u>ubiquitous</u> and include the following examples

Practical applications

- Internet world
- Electronic commerce
- Manufacturing and other commercial settings
- Shortest path
- Matrices multiplication order
- DNA sequence matching

Common about algorithms

- There are many candidate solutions, most of which are not what we want, finding one that we do want can present quite a challenge.
- There are practical applications (its not just mathematical exercises to develop algorithms.)

Why Study Algorithms and Performance of Algorithms

- Algorithms help us to understand scalability.
- Performance often draws the line between what is feasible and what is impossible.
- Algorithmic mathematics provides a language for talking about program behavior.
- Performance is the currency of computing.
- The lessons of program performance generalize to other computing resources.
- Speed is fun!

What is Abstract Data Type

A definition for a data type solely in terms of

- a set of values and a
- set of operations on that data type.

The definition consists of:

- **storage structures (data structures) to store the data items**

  and

- **algorithms for the basic operations.**

What is a Data Structure

- A data structure is a way to store and organize data in order to facilitate access and modifications.
- No single data structure works well for all purposes
- Need to know the strengths and limitations of several of them.

Technique

- Can’t get a “cookbook” for algorithms?
- Many problems you will encounter don’t have any published algorithm.
- So need to learn “techniques” of algorithms design and analysis
- So you develop algorithms in your own, show that they give correct answer and understand their efficiency.
- We will learn several such techniques in later part of this course.

Algorithms and other technologies

- Is an algorithm a technology like hardware, etc?
- Total system performance depends on choosing “efficient” algorithms as much as choosing fast hardware.

Algorithms and other advanced technologies

- Hardware with high clock rates, pipelining and superscalar architecture.
- Easy to use graphical user interface (GUI's)
- Object oriented systems.
- Local-area and wide-area networking.

- Are algorithms as important as above technologies?

Course-Overview

- Basics of Algorithms
- Asymptotic Notations
- Sorting Algorithms
- Divide and Conquer
- String Matching Algorithms
- Graph Theory
- Dynamic Programming
- NP complete problems

Psuedocode

- Pseudocode is a compact and informal high-level description of a program using the conventions of a programming language, but intended more for humans.
- Pseudocode is not an actual programming language.
- So it cannot be compiled into an executable program.

Psuedocode

- **INPUT** – indicates a user will be inputting something
- **OUTPUT** – indicates that an output will appear on the screen
- **WHILE** – a loop (iteration that has a condition at the beginning)
- **FOR** – a counting loop (iteration)
- **REPEAT – UNTIL** – a loop (iteration) that has a condition at the end
- **IF – THEN – ELSE** – a decision (selection) in which a choice is made any instructions that occur inside a selection or iteration are usually indented

Psuedocode Example

- Calculate Area and Perimeter of Rectangle

```
BEGIN
NUMBER b1,b2,area,perimeter
INPUT b1
INPUT b2
area=b1*b2
perimeter=2*(b1+b2)
OUTPUT area
OUTPUT perimeter
END
```

Pseudocode If Else Example

```
BEGIN
NUMBER age

INPUT "Enter your age for driving licence"
OUTPUT age

IF age>=18  THEN
    OUTPUT "You can take driving licence"
ELSE
  OUTPUT "You can't take driving licence"
ENDIF
END
```

Pseudocode For Loop Example)

```
BEGIN
NUMBER counter

FOR  counter = 1 TO  100 STEP 1 DO
    OUTPUT counter
ENDFOR

END
```

Read 10 numbers and find sum of even numbers

```

BEGIN
NUMBER counter, sum=0, num
FOR counter=1 TO 10 STEP 1 DO
    OUTPUT "Enter a Number"
    INPUT num
    IF num % 2 == 0 THEN
        sum=sum+num
    ENDIF
ENDFOR
OUTPUT sum

END
```

Find the sum of all elements of array

```
PROCEDURE SUM_Array
BEGIN
Input:List numbers[1..n], n
Output: Sum of an array
sum=0
FOR i=0 to n-1
  sum = sum + numbers[i]
ENDFOR

OUTPUT  "Sum of numbers in the array"+sum

END
```

Reference

- Chapter#1 (The Role of Algorithms in Computing)
- Introduction to Algorithms
- Thomas H.Corman